

MARTIN MOKOTLANA MAMADISE

Computer Engineering Graduate | Software Development | Systems & Networking | Technical Support
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PROFESSIONAL SUMMARY

Computer Engineering graduate and current Honours student with hands-on academic experience in technical troubleshooting, networking, software testing, API testing and SQL. Strong analytical and problem-solving skills with experience investigating system faults, documenting technical findings and working across Linux and Windows environments. Eager to develop practical experience in production support, incident management and customer-focused technical operations.

KEY SKILLS

- TCP/IP, VLANs, Routing & Switching
- Network Security Fundamentals (Firewalls, VPNs)
- Network Simulation: GNS3, Cisco Packet Tracer, Wireshark
- Linux & Windows Administration (CLI, system config)
- Systematic Troubleshooting & Root-Cause Analysis
- Technical Documentation & Incident Logging
- Python Scripting (network automation & analysis)
- Fast Learner - comfortable with new tools and standards

TECHNICAL COMPETENCIES

- **Operating Systems:** Linux (CLI proficiency, Windows, VirtualBox (VM configuration & management)
- **Security:** Firewall concepts, Access Control Lists (ACLs), VPN fundamentals, network policy review
- **Networking:** TCP/IP, OSI Model, IP Addressing, VLANs, DNS, Routing Protocols, GNS3, VyOS, Cisco, Packet Tracer, Wireshark.
- **Programming/Scripting:** Python, C, Java, basic JavaScript; Git/GitHub for version control
- **Databases:** MySQL - querying and data validation
- **Development Tools:** Wireshark (packet analysis), VS Code, Git/GitHub, Apache NetBeans, Postman (API testing)
- **Embedded/IoT Systems:** ESP32, Arduino - sensor integration and system diagnostics
- **Technical Support:** Troubleshooting, fault isolation, root-cause analysis, incident documentation
- **Software & Testing:** Postman, REST/API testing, application debugging, functional testing

RELEVANT PROJECT EXPERIENCE

Network Virtualization & Routing Simulations

2025

Cape Peninsula University of Technology

- Built and configured multi-router virtual network topologies using GNS3 and VyOS, simulating enterprise-level environments with VLANs, routing protocols, and segmented subnets
- Implemented and verified inter-router routing, troubleshooting end-to-end connectivity failures through systematic packet tracing and configuration review - directly analogous to firewall rule debugging
- Documented lab findings and configuration steps in structured reports, building the habit of clear technical record-keeping

Smart Agriculture IoT Monitoring System

2025

Cape Peninsula University of Technology

- Built an ESP32-based IoT system integrating multiple sensors for real-time data collection and remote monitoring

- Diagnosed and resolved hardware/software integration issues through iterative testing and root-cause analysis
- Documented system behaviour and improved reliability through structured troubleshooting - mirrors service desk incident tracking

Traffic Engineering & Network Optimisation

2025

Cape Peninsula University of Technology

- Developed a Python-based network optimization tool simulating large-scale traffic routing and congestion scenarios across a multi-link topology
- Reduced maximum link utilization from over 600% to under 75% through iterative analysis and targeted routing adjustments - demonstrating security-relevant skills in traffic monitoring and anomaly identification
- Applied structured root-cause analysis to identify bottlenecks, mirroring the incident investigation process in security operations

Brick Breaker WebGL Game

2024

Cape Peninsula University of Technology

- Debugged complex logic and performance issues in a real-time browser-based application
- Documented and resolved rendering and collision detection faults through methodical testing

EDUCATION

Bachelor of Engineering Technology in Computer Engineering (Honours) 2026 - In Progress

Cape Peninsula University of Technology | Bellville Campus

Bachelor of Engineering Technology in Computer Engineering

2023 - 2025

Cape Peninsula University of Technology | Bellville Campus

National Senior Certificate (Matric)

2022

Kopa Secondary School